

2.3 Codecs

By the mid '90s, usage of internet increased manifold and with the dawn of the new millennium certain radical changes were happening in the world of digital audio. Audio CDs, by far, were in charge and everybody and anybody had a CD player. Nonetheless, there was one fundamental problem. First, CD's scratch easily and second, one cannot transport a CD over the net physically.

With the Internet slowly becoming the hub of all entertainment, it was necessary to look for other, more cost effective options for a good audio-visual experience that did not hog bandwidth yet provided decent quality-both in terms of audio and video. This was the basic idea that facilitated the inception of audio and video codecs.

A "codec" can be defined as the combination of two words-Encoder and Decoder. These encoders and decoders can be either hardware or software, but we will only talk about the software codecs.

Before we actually delve into the world of software codecs, let us get our basics on digital music right.

When we talk about a software codec, it is a software program containing an algorithm to compress and then decompress data. For instance, if an MP3 file is encoded at a bit-rate of 128K from a source file that is 20MB in size (uncompressed), then the resulting file would be approximately 2MB in the compressed form, going by the 11:1 compression ratio, which the MP3 codec compresses at.

Talking about compression ratios, it is the ratio of the size of the original uncompressed audio file and the compressed resulting audio file.

Next up are the two categories of codecs-lossless and lossy respectively. An example of a lossless codec is WMA and a lossy codec is MP3. The term "lossless" means that the compressed file is